AUTOMATION EXERCISE WEBSITE

Information Technology Institute

Web Testing Project

Quality Assurance — Graduation Project

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48	41	7	85.4%
Test Cases	Passed	Defects	Pass Rate

AGENDA

01 **Project Overview**

What is Automation Exercise? Goals & Scope

02 **Technology Stack**

Manual (Excel · Chrome) vs Automation (Java · Selenium · TestNG · Maven · Allure)

03 **Tool Comparison**

Why we chose these tools over the alternatives

04 **Test Plan & Strategy**

Test Plan highlights · BVA · EP · Negative Testing · 5 Code Modules

05 **Test Coverage & Execution**

48 test cases across 5 functional modules — full breakdown

06 **Defects Found**

7 defects with root causes

07 **Test Plan Highlights**

Entry/exit criteria, risks, recommendations

08 **Conclusion**

Results summary & next steps

01 | PROJECT OVERVIEW

What is Automation Exercise?

automationexercise.com
is a fully functional e-commerce web application designed specifically for QA automation practice. It provides a realistic environment to test:

- User Registration & Account Management
- Product Browsing & Shopping Cart
- Search & Filter Functionality
- Subscription
- Login, Logout & Delete Account

Project Goal

Design and execute a comprehensive test suite covering all key user journeys — validating correct behavior, catching input-validation defects, and ensuring production-readiness.

In Scope

- Registration / Signup (2 steps)
- Login · Logout · Delete Account
- Cart: Add · Remove · Quantity
- Subscription Email Validation
- Product Search

02 | TECHNOLOGY STACK

🖊️ Manual Testing

Excel Sheet

Test Management

Human-readable test case repo — TC ID, steps, expected vs actual, pass/fail. Easy to share and review.

Chrome Browser

Test Environment

All manual tests executed on Chrome for Windows — the most widely used browser, ensuring real-world coverage.

⚙️ Automation Testing

Java

Test Language

Strongly typed, OOP support for Page Object Model, industry standard for Selenium.

Selenium

Browser Automation

Simulates real user interactions — clicks, typing, form submission across browsers.

TestNG

Test Framework

Parallel execution, grouping, data providers, and rich annotations (@BeforeSuite etc.).

Maven

Build & Dependency

One POM file manages all dependencies — reproducible builds, no JAR conflicts.

Allure

Reporting

Interactive HTML report: step-by-step logs, screenshots on failure, trends dashboard.

03 | WHY THESE TOOLS — NOT THE ALTERNATIVES

Java vs Python

✓ We used: Java

- Industry standard for Selenium
- Strong OOP support for Page Object Model
- Excellent IDE support (IntelliJ)
- Strong typing catches errors early

✗ Not: Python

- Dynamic typing leads to runtime errors
- Less common in enterprise QA teams
- Weaker TestNG/Maven integration

TestNG vs JUnit

✓ We used: TestNG

- Built-in data providers for DDT
- Test grouping & dependency support
- Parallel test execution out of box
- Richer annotations (@BeforeSuite etc)

✗ Not: JUnit

- No built-in parallel execution
- Weaker test grouping support
- Requires Parameterized extension for DDT

Selenium vs Playwright

✓ We used: Selenium

- Industry standard for web automation
- Huge ecosystem & community support
- Excellent Page Object Model support
- Mature integration with Java, TestNG & Maven

✗ Not: Playwright

- Newer ecosystem compared to Selenium
- Less common in existing enterprise Java QA stacks
- Smaller pool of Selenium-style learning resources

04 | TEST PLAN & STRATEGY

Environment:	Chrome · Windows · automationexercise.com	**Entry Criteria:**	App live · Selenium configured · TCs approved
Exit Criteria:	All 48 TCs executed · Defects logged · Pass rate ≥ 80%	**Code Modules:**	Register · confirmationUser · Products · deleteAccount · validationErrorMessage

EP

Equivalence Partitioning

Divided inputs into valid/invalid classes. One value per partition tested — e.g. valid email, empty email, email missing @.

BVA

Boundary Value Analysis

Tested edges of valid ranges — empty fields, spaces-only, max-length inputs — where most bugs hide.

NEG

Negative Testing

Systematically fed invalid data to every field: special chars, Arabic names, alphabetic phone numbers, duplicate emails.

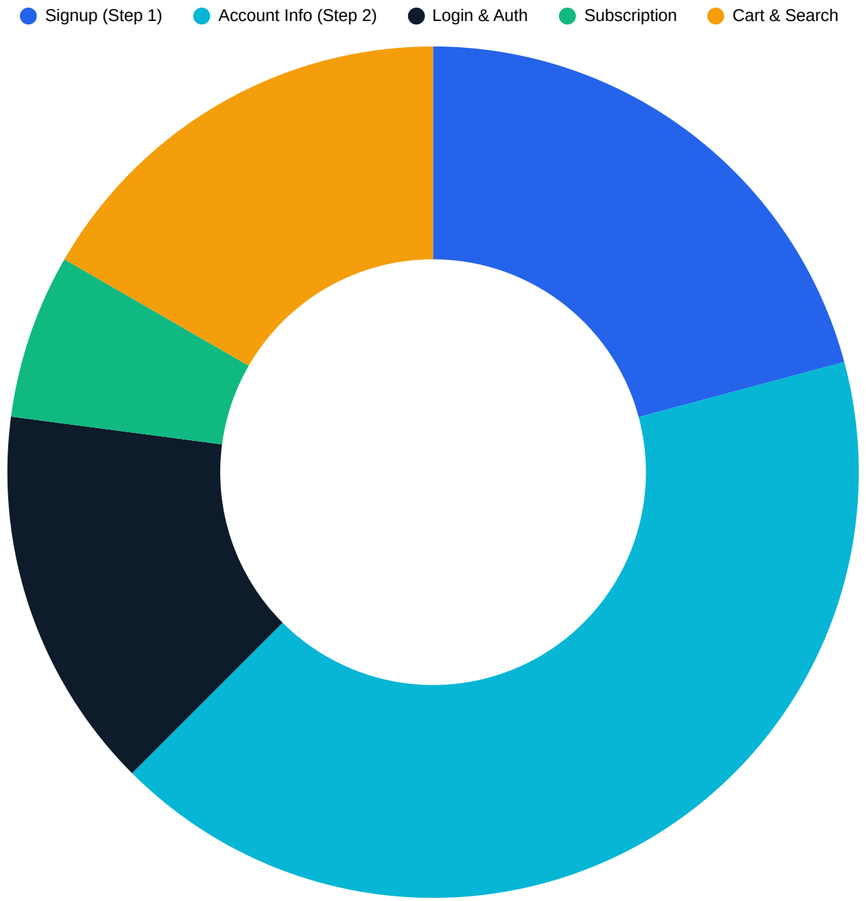
E2E

End-to-End Testing

Full user journeys: Signup Step 1 → Account Info → Login → Browse → Cart. No isolated unit tests.

Test Levels: System Testing (integrated app) · Functional Testing · Regression Checks per Module

05 | TEST Case Design



Module	TCs	Pass	Fail
Signup — Step 1	10	9	1
Account Info — Step 2	20	14	6
Login & Auth	7	7	0
Subscription	3	3	0
Cart & Search	8	8	0
TOTAL	**48**	**41**	**7**

A total of 48 test cases covering positive & negative scenarios across 5 API modules.

06 | TEST PLAN HIGHLIGHTS

Entry & Exit Criteria

Entry (before testing begins)

- App accessible at automationexercise.com
- Chrome + Selenium configured
- All TCs reviewed & approved
- Test data prepared

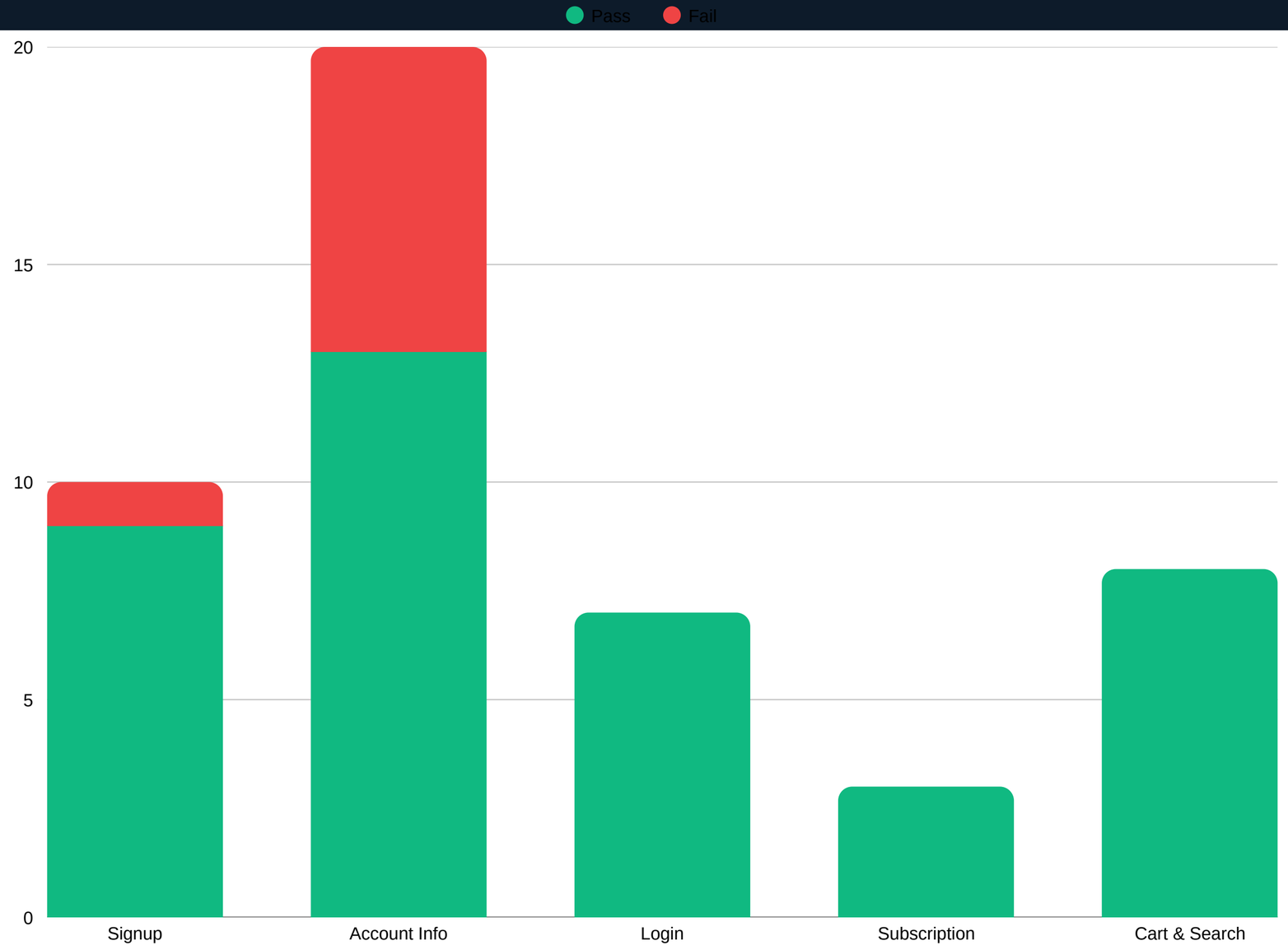
Exit (testing complete when…)

- All 48 TCs executed
- All defects logged with severity
- Pass rate ≥ 80% achieved ✓ (85.4%)
- Test Summary Report ready

Recommendations

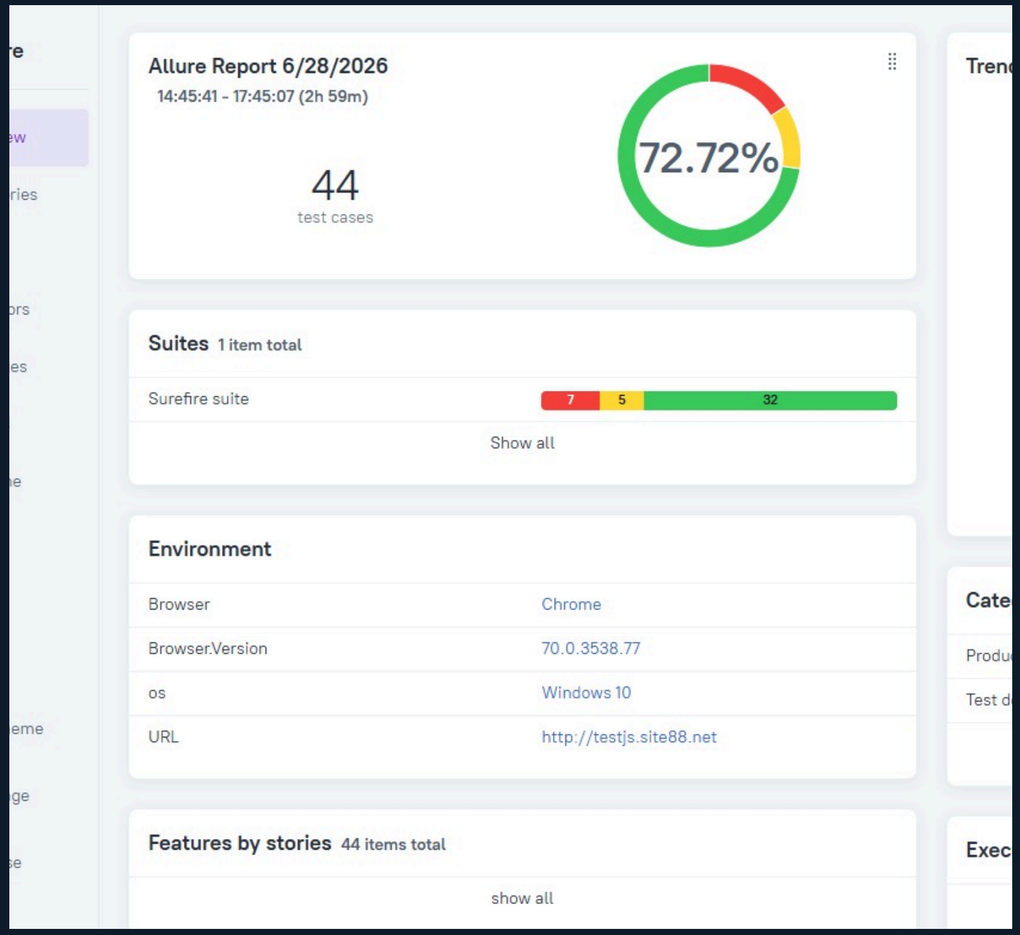
01	Enforce phone validation — digits only + optional leading +	Critical
02	Sanitize name fields — reject spaces-only & special-char inputs	Critical
03	Add server-side validation — all current validation is client-side only	High
04	Expand to Firefox, Edge, Safari — Chrome-only is insufficient for release	Medium

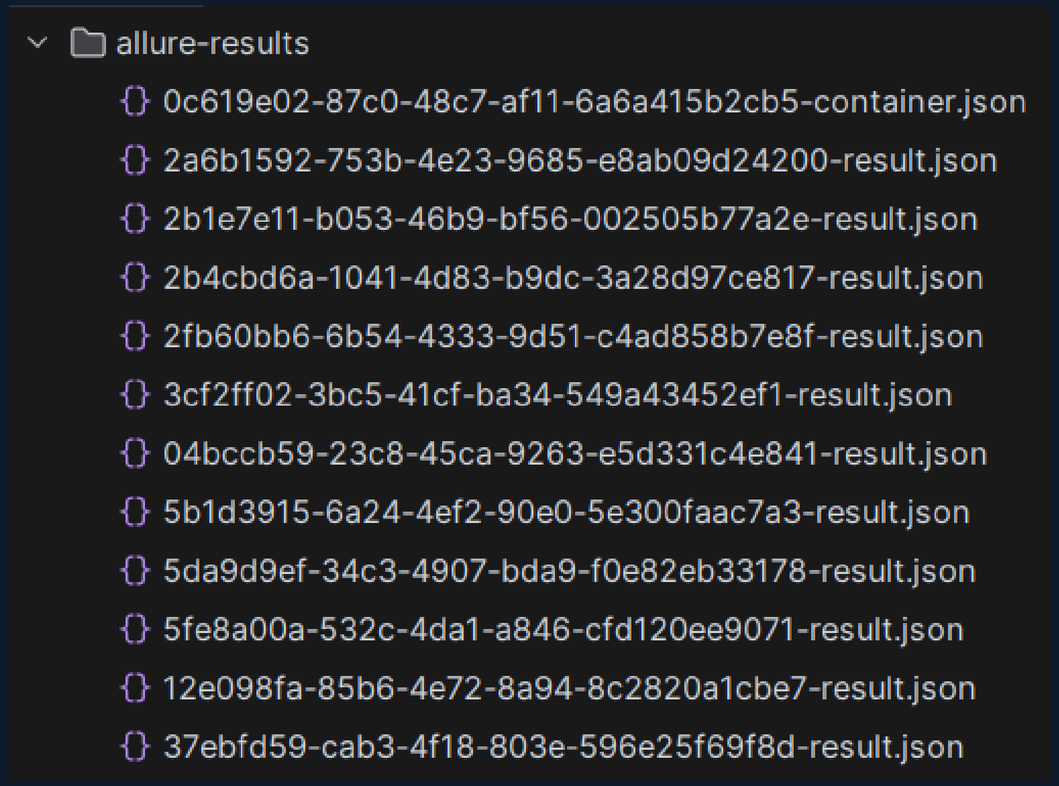
Test Results

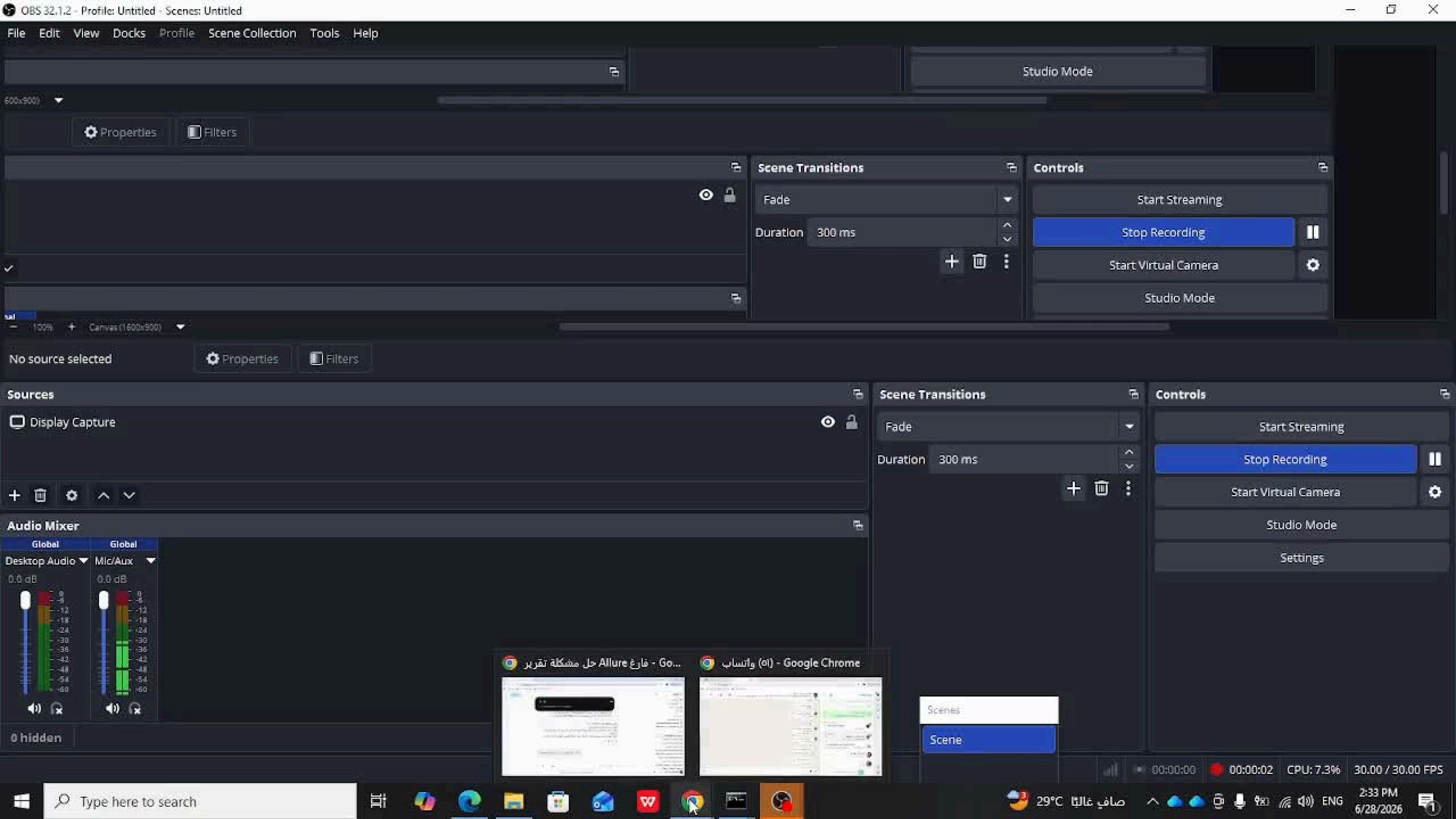


48 Total TCs	**48** Executed
41 Passed	**7** Failed
85.4% Pass Rate	**3** Critical Bugs
4 High Bugs	**100%** Login/Cart/Sub

+ Allure Report Overview



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Properties
Filters
Scene Transitions
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Duration 300 ms
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CONCLUSION

Solid foundation. One fixable gap.

- 85.4% pass rate — all core flows (Login, Cart, Search, Subscription) are production-ready.
- 7 defects share one root cause: missing input sanitization on name & phone fields — one fix pattern closes all bugs.
- Selenium + Java + TestNG + Maven stack delivered reliable, repeatable, reportable automation.
- Allure Report gave us a professional interactive dashboard — every failure documented with full steps and actual result.

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+ GitHub Repo

Thank you so much

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Test Case Report